

Network Operations Strategy Memo

TO: Head of Network Operations, Delhivery

FROM: Data Science & Strategy Team

DATE: June 2026

RE: Graph-Based Network Intelligence - Bottleneck Analysis & Action Plan

Situation:

Analysis of 142,502 delivery segments across 1,657 hubs over 22 days reveals a concentrated problem: **five hubs are responsible for 70.6% of all SLA damage across the network**. Of 118,708 SLA-bound trips, 99,512 (83.8%) missed their delivery deadline. This is not a random distribution - it is structurally concentrated and actionable.

A second finding compounds the problem: OSRM, the routing engine Delhivery uses to estimate delivery times, underestimates actual delivery time by an average of 2.1x across all corridors. 73.7% of all corridors are chronically delayed - meaning the gap between promised and actual delivery is systemic, not isolated.

The Five Hubs Requiring Immediate Attention:

Our analysis combines structural graph analysis (which hubs are chokepoints in the network) with SLA breach contribution (which hubs cause the most actual damage) to produce a bottleneck score for all 1,657 hubs.

Rank	Hub	SLA Breach Rate	Network Risk Share	Intervention
#1	Gurgaon Bilaspur HB	81.1%	38.5%	Parallel route
#2	Bangalore Nelmngla H	81.9%	16.2%	Route-type shift
#3	Kolkata Dankuni HB	84.9%	3.0%	Facility upgrade
#4	Hyderabad Shamshabad H	88.6%	2.3%	Facility upgrade
#5	Bhiwandi Mankoli HB	90.1%	10.7%	Parallel route

Gurgaon alone accounts for 38.5% of network-wide SLA damage. With betweenness centrality of 0.233 - the highest in the network - it sits on the shortest path between more hub pairs than any other facility. Delays here cascade across the entire network.

Recommended Interventions:

Hub #1: Gurgaon Bilaspur HB: Parallel Route

Gurgaon's breach rate (81.1%) is already among the best-performing large hubs in the network. The problem is volume, not operations. 22,275 SLA-bound trips pass through this hub in 22 days - more than double any other hub. **Recommendation:** Create parallel inter-city corridors via Manesar and Faridabad to redistribute 20-30% of Gurgaon's FTL volume. This does not require facility investment - only route planning.

Hub #2: Bangalore Nelmngla H: Route-Type Shift

Bangalore's three worst outgoing corridors are all internal city hops (KH Road 2.33x, Whitefield 2.0x, Peenya 1.88x). A national sorting hub is being asked to do last-mile city delivery. **Recommendation:** Shift Bengaluru city corridors from FTL to Carting. Our FTL vs Carting framework confirms Bangalore as a clear Carting candidate for city hops - breach rate drops significantly. Estimated SLA breach reduction for city corridors: 15-20%.

Hub #3: Kolkata Dankuni HB: Facility Upgrade

Kolkata's incoming feeder routes are catastrophically delayed - Ranaghat (11.6x), Helencha (11.6x), Midnapore (7.5x). These are not road congestion issues - factors above 10x indicate facility dwell time at Kolkata itself. **Recommendation:** Increase unloading dock capacity and staffing at Dankuni. Night breach rate (85.5%) is marginally worse than day (84.6%), suggesting a scheduling gap during evening receiving windows.

Hub #4: Hyderabad Shamshabad H: Facility Upgrade

Night breach rate of 90.1% vs day rate of 87.1% is a clear night-shift signal. Incoming corridors from Tolichowki (2.64x) and Uppal (2.64x) are delayed regardless of time of day - the bottleneck is inside the facility. **Recommendation:** Add night-shift dock staff and extend facility operating hours to 24/7. Expected breach rate improvement: 3-5 percentage points within 90 days.

Hub #5: Bhiwandi Mankoli HB: Parallel Route

Bhiwandi receives high-volume Mumbai feeders — Mumbai Hub (171 trips, 2.55x), Chandivali (238 trips, 2.48x), Mira Road outgoing (148 trips, 2.44x). The 90.1% breach rate is the highest among the top 5. **Recommendation:** Create direct Mumbai to destination corridors bypassing Bhiwandi for the highest-volume flows. This decongests Bhiwandi and reduces transit hops for Mumbai shipments simultaneously.

Impact If Top 3 Hubs Are Upgraded:

Upgrading Gurgaon, Bangalore, and Kolkata to the performance level of Delhivery's top-10% hubs (68.9% breach rate):

- **4,266 fewer SLA breaches** in an equivalent 22-day window
- **4.3% reduction in late deliveries** network-wide
- **8.9% of total SLA risk recovered** (2,776,266 cutoff points)

The 4.3% figure is honest - Gurgaon and Bangalore are already well-operated. The gain comes from route diversification reducing absolute volume at these hubs, not from fixing poor operations. Kolkata has the most direct operational improvement headroom.

Smarter ETA Predictions - What Changes:

Our graph-enhanced ETA model reduces prediction error by 30.8%:

Model	MAE	Within 15% of actual
Current OSRM baseline	55.45 min	50.5% of trips
Graph-enhanced model	38.34 min	65.0% of trips

65% of delivery promises will now be within 15% of actual delivery time, up from 50.5%. The model learns corridor-specific delay patterns - it knows that Gurgaon to Kanpur runs at 1.85x OSRM, not the network average.

Analysis based on 142,502 trip segments, Sep 12 - Oct 3 2018. Graph model uses node2vec embeddings on a 1,657-node directed network. FTL vs Carting framework based on 68 hubs with observed data for both route types. All findings reproducible from the accompanying codebase.